

flashjet

GPU-native jet clustering with the full merge history

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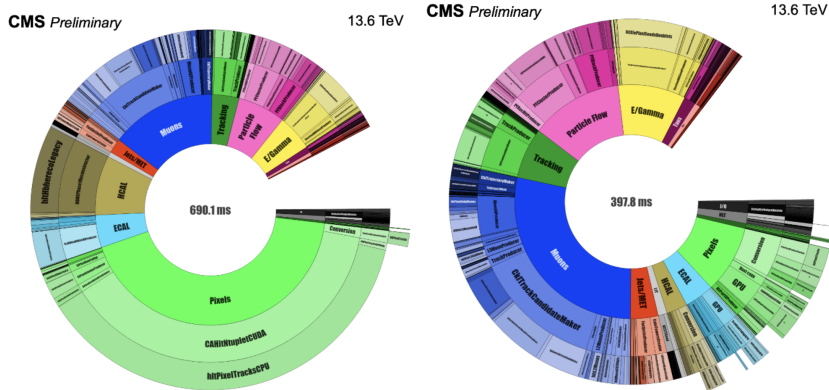
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- 1 Jet clustering, and flashjet
- 2 Correctness
- 3 Speed
- 4 Downstream: the clustering tree as tagger input

Jet clustering, and flashjet

Reconstruction is moving to accelerators — clustering has not



CMS HLT per-event time, CPU-only (left) vs with GPU offload (right) — $\sim 30\%$ of the reco offloaded buys $\sim 40\%$ less HLT time. **Jets/MET** barely moves: it is still on the CPU. [CMS, EPJ Web Conf. 295 (2024) 11020]

Why clustering stayed on the CPU

- sequential recombination is **IRC safe** — that is why it won
- but each merge **depends on the previous one** — serial dependence is what a GPU does worst

What it does

- generalised- k_t — anti- k_t , k_t , C/A
 - **many jets or events per launch**
 - returns jets, the **full merge history**, and substructure **as tensors**
 - written in **Triton**: autotunes for whatever GPU it finds — no separate CUDA build.
- ⇒ built for **reclustering inside training loops**, on padded $(B, N, 4)$ tensors that **stay on the GPU**

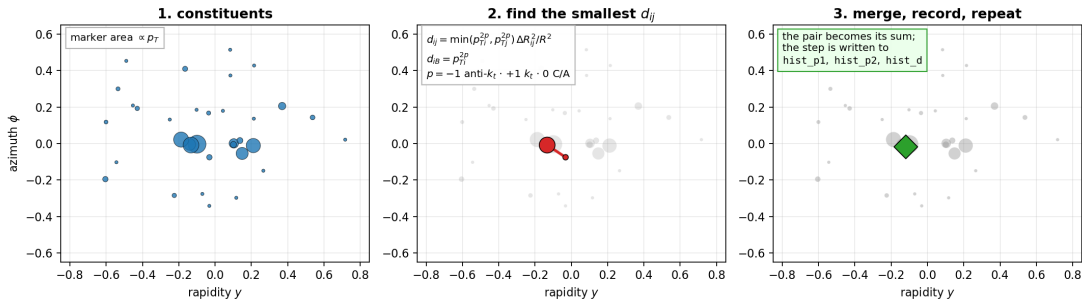
Why it can be done at all

- Cacciari–Salam lemma: the smallest d_{ij} in the whole event is always some particle's **geometric** nearest neighbour
 - so each particle tracks only **its own** NN, and one merge invalidates $O(1)$ of them
- ⇒ $O(N^2)$, **not** $O(N^3)$ — the serial chain stays, but each step is cheap and parallel

In the ML ecosystem

- a **PyTorch** call — tensors in, tensors out
- no host $\leftrightarrow$ device copy in a training loop

Sequential recombination — and what flashjet keeps from it



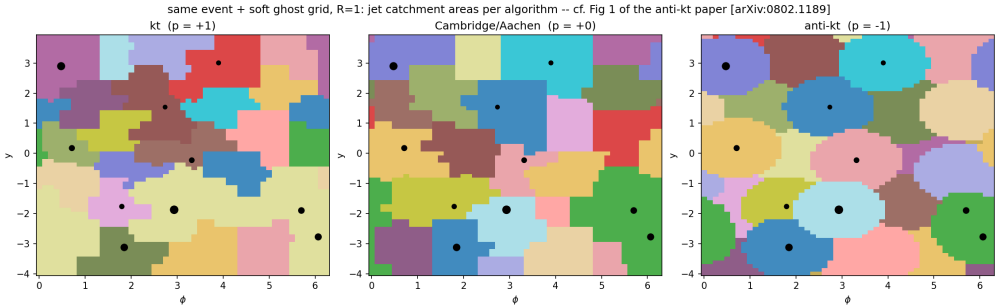
Two distances, computed for every pair:

- d_{ij} — *pair* distance, particle i to particle j
- d_{iB} — *beam* distance, particle i to the beam

Then repeat until nothing is left:

- ① find the smallest of all distances
- ② if it is a d_{ij} : **merge** i and j
- ③ if it is a d_{iB} : call i a **jet**, remove it

Both distances were defined on the previous slide; the exponent p there selects which algorithm this loop runs.



Only the exponent p changes:

p	algorithm	merges first	tree ordered by	used for	panel above
-1	anti- k_t	the hardest pairs	—	finding jets	right — circular, rigid
0	Cambridge–Aachen	the closest pair	angle	grooming, Lund plane	middle — ragged edges
+1	k_t	the softest pair	relative p_T	exclusive subjets	left — most irregular

Only anti- k_t gives jets whose area is insensitive to the soft background — which is why it is the one used to *find* jets. The other two are used on a jet that has *already* been found.

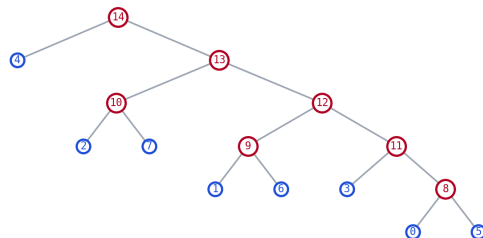
Clustering a batch returns **two** things — the jets ($\text{jet_idx}(B, N)$, $\text{n_jets}(B, N)$), and the **merge history** below: hist_p1 , hist_p2 (which pair merged), hist_child (what they merged into) and hist_d (the distance), each (B, N) .

what the kernel writes out (step $\rightarrow$)

hist_p1	0	1	2	3	9	10	4
hist_p2	5	6	7	8	11	12	13
hist_child	8	9	10	11	12	13	14
hist_d	0.000	0.000	0.000	0.029	0.030	0.053	0.059
step	0	1	2	3	4	5	6

each column is one merge: particles p_1, p_2 join into *child*, at distance d . Indices ≥ 8 are merged pseudo-particles.

the binary tree those rows encode



blue = the 8 input particles red = merged pseudo-particles

One real ATLAS jet with 8 constituents, C/A. Read column 0: particles 0 and 5 merge into 8 — the node at the bottom right. Seven columns, seven merges, and the tree is complete. **Everything downstream — grooming, subjects, Lund coordinates — is a read of these rows, never a second clustering.**

What is implemented

Clustering

- generalised- k_t : anti- k_t , k_t , C/A
- merge history recorded in-kernel
- many jets clustered in one batched call
- scales from single jets to whole events

Substructure — reads of the history

- **F1** exclusive k_t subjects
- **F2** soft drop / mMDT / mass-drop
- **F3** primary Lund coordinates

All three are pure-torch post-reads: no kernel changes, CPU/GPU identical, negligible cost.

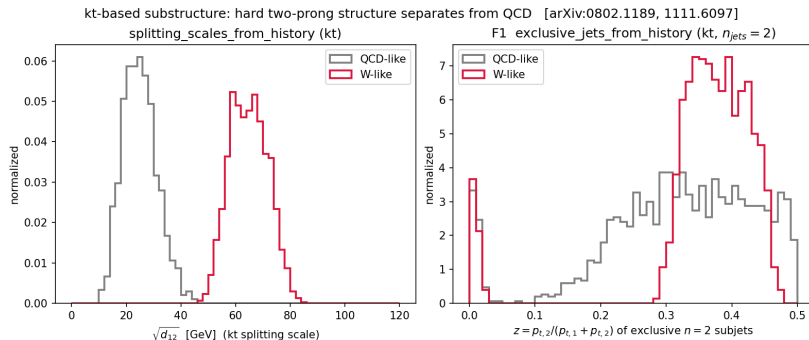
What it does. Undo the last merges of the sequence to expose exactly n subjects — or stop at a d_{cut} .

How. The history is already a binary tree. Walk back up it $n - 1$ steps; no reclustering.

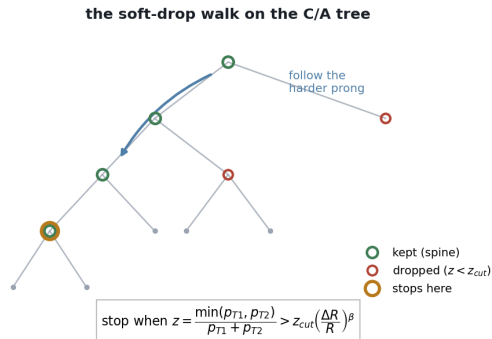
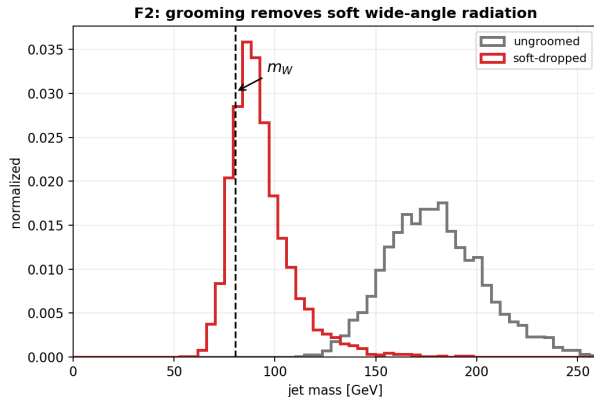
Why k_t . k_t merges the softest pair first, so the *last* merges are the hard prongs — exactly what a 2-prong decay leaves behind.

The splitting scales come straight from `hist_d`:

$$\sqrt{d_{12}}, \sqrt{d_{23}}, \dots$$



Toy W-like (2-prong) vs QCD-like fat jets. **Left:** the k_t splitting scale $\sqrt{d_{12}}$ separates cleanly. **Right:** the momentum balance z of the two exclusive subjects — the W sits near $z \sim 0.4$, QCD spreads to small z .

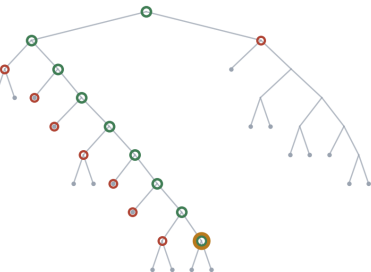


Walk down the C/A tree along the harder branch, discarding the softer prong until that condition is met. $\beta = 0$ recovers mMDT.

The walk is $O(\log_2 n)$ on the stored tree — it never touches the constituents again, so grooming is essentially free once clustered.

vertical = declustering depth · horizontal = angular ordering

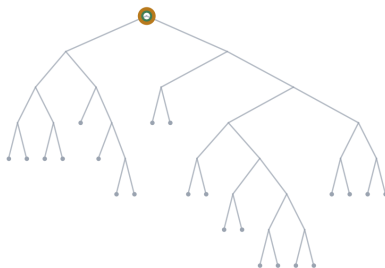
Light quark / gluon jet



a long spine: soft prong after soft prong
is stripped, and the mass collapses

$n_{drop} = 8$ $m: 80 \rightarrow 1 \text{ GeV}$

Boosted top — a clean two-prong core



the very first split is already balanced,
so nothing is groomed away

$n_{drop} = 0$ $m: 80 \rightarrow 80 \text{ GeV}$

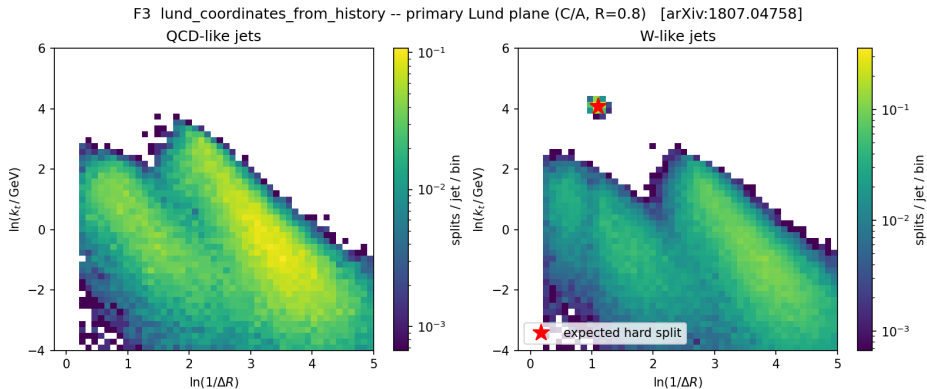
Boosted top — the full $t \rightarrow bW$ system



also stops at once, but on a wider split,
so the whole decay is kept

$n_{drop} = 0$ $m: 170 \rightarrow 170 \text{ GeV}$

● soft-drop spine ● dropped prong ● grooming stops here ● constituent



What. Every primary splitting emits $z = \frac{p_{T2}}{p_{T1} + p_{T2}}$, ΔR , $k_t = p_{T2} \Delta R$, and the plane coordinates $(\ln \frac{1}{\Delta R}, \ln k_t)$.

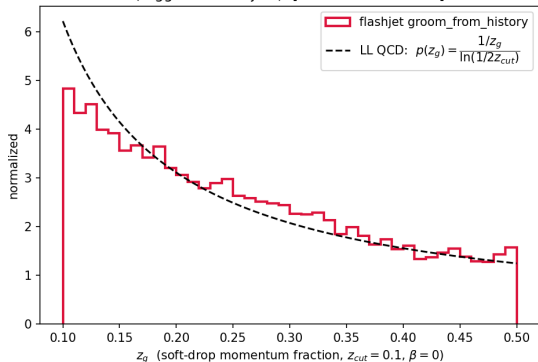
QCD fills the soft-collinear region smoothly; the W-like sample adds a **localised hard splitting** exactly where the decay is predicted to sit (star).

How. One pass down the primary branch — each node is one point in the plane.

Check. The d -channel ties *exactly* to the splitting scales computed independently.

Correctness

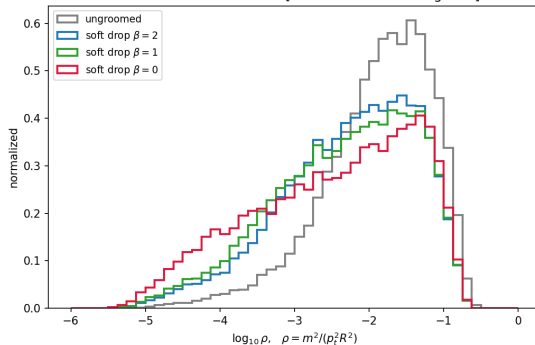
groomed splitting fraction vs the analytic prediction
(tagged 72% of jets) [cf. arXiv:1402.2657]



Soft-drop z_g follows the $1/z$ form predicted at leading-log, recovered from the C/A history.

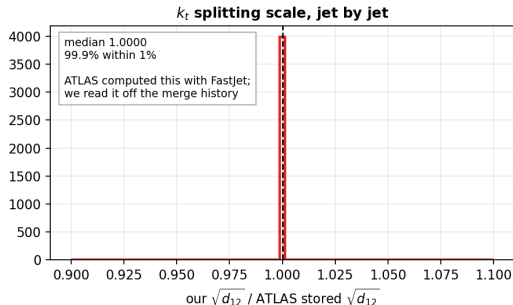
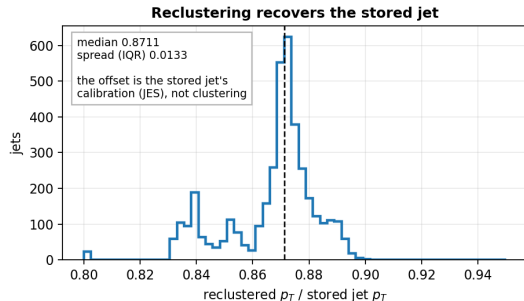
Toys first: on generated input the answer is known analytically, so anything off the curve is a real bug. Real-data checks follow. Toy-shower input tests the decoders, not an event generator.

groomed jet mass: stronger grooming (smaller β) pushes the soft-radiation mass down [cf. arXiv:1402.2657 Figs 3-4]



Groomed-mass ordering in β : larger β grooms less, so the mass peak moves up.

Jet-by-jet closure against ATLAS-stored quantities



Left. Reclustering the stored constituents recovers the stored jet with a **0.013 spread**. The 0.87 offset is the stored jet's **calibration**, applied after clustering — it is present before we touch anything.

Right. The k_t splitting scale $\sqrt{d_{12}}$ against the value **ATLAS computed with FastJet** and stored: median **1.0000**, **99.9 %** within 1%. We read ours straight off the merge history.

vs FastJet — a 150-point grid

3 algorithms $\times$ 5 radii $\times$ 5 multiplicity bins $\times$ 2 samples,
20 000 jets each.

100.000 % jet-count agreement at **every** point.

Leading-jet p_T within 10^{-4} at 148/150; median difference
 $\sim 7 \times 10^{-8}$ — float32 precision.

vs ATLAS's own reconstructed jets

Cluster the ~ 600 stored calorimeter clusters
ourselves; compare to the jets ATLAS reconstructed
and published.

100.0 % n_{jets} match, **10.96 vs 10.96** jets/event.

The event regime, where the problem is $O(N^2)$.

Two *independent* references: FastJet for the jets, a separate **NumPy tree-walk** for every substructure decoder.
The second test closes the algorithm against **the experiment's published output** — not against another
implementation of the same algorithm.

Per-point tables and the R /multiplicity scans are in backup.

Speed

Two ways the clusterer gets used

Jet regime — one *jet* per unit

Recluster a single jet's constituents to read substructure off the tree. **Small N , huge batch** — thousands of independent jets per launch. This is the ML use case: features inside a training loop.

Event regime — one *event* per unit

Cluster a whole event's calorimeter clusters to *find* the jets. **Large N , modest batch** — the classic reconstruction task.

Same kernel, same API — only the shape of the input tensor differs. The regimes are timed separately because a jet and an event are different units.

Data — ATLAS Open Data, freely presentable

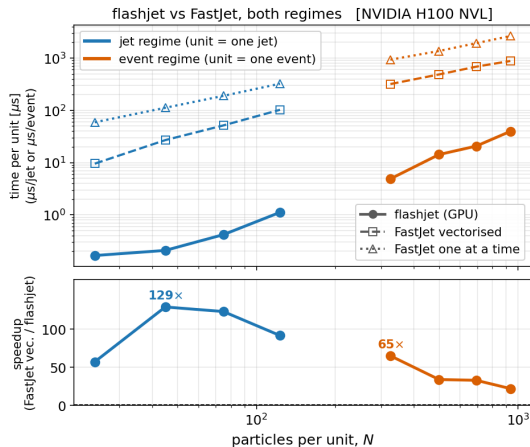
	sample	units	$\langle N \rangle$
jet	boosted top	30 000	59.2
jet	QCD (q/g)	30 000	52.4
event	$t\bar{t}$ reco	3 000	591.6

Grids: **3 algorithms** $\times$ **5 radii** $\times$ **5 multiplicity bins** in each regime — 150 jet points, 75 event points.

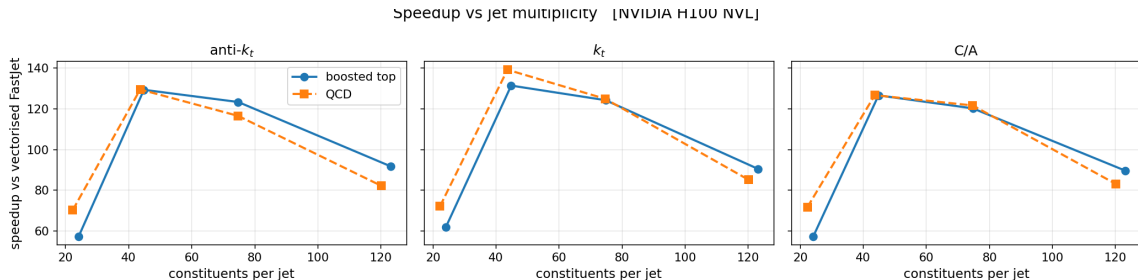
Method (identical in both)

- 1 extract constituents once, save;
- 2 cluster on GPU, time it;
- 3 cluster the *same saved file* with FastJet;
- 4 **check they agree** — else the timing is meaningless.

Speedups are quoted against the **vectorised** FastJet interface — the faster, fairer baseline.



Colour = regime, style = implementation. The unit is a *jet* (blue) or an *event* (orange) — only the **ratio panel** compares across them. **0.17–1.1 $\mu\text{s}/\text{jet}$, 4.9–40 $\mu\text{s}/\text{event}$** ; speedup peaks **129 $\times$ / 65 $\times$** .



All three algorithms, both samples, $R = 1.0$. **54–143 $\times$** over vectorised FastJet across the 150-point grid (median **89 $\times$**), every point at **100 %** n_{jets} agreement. The speedup **peaks near** $N \approx 45$: below it the GPU is under-filled, above it the $O(N^2)$ work per jet grows faster than FastJet's. The peak sits squarely in the **boosted-object range**.

Using it

```
import flashjet

out = flashjet.cluster(p4, mask, R=1.0, algorithm="antikt")
```

argument	type	
p4	$(B, N, 4)$ tensor	(p_x, p_y, p_z, E) ; CPU or CUDA
mask	(B, N) bool	marks real constituents in a padded batch
R	float	jet radius
algorithm	str	"antikt" "kt" "cambridge"
p	float	generalised- k_t exponent (overrides algorithm)

out.n_jets	# (B,)	jets found
out.jets_p4(p4)	# (B,J,4)	jet four-momenta
out.splitting_scales()	#	$\sqrt{d_{12}}$, $\sqrt{d_{23}}$, ...
out.exclusive_jets(n_jets=3)	# F1	exclusive k_t subjects
out.groomed_jets(p4, R, z_cut=0.1, beta=0.0)	# F2	soft drop / mMDT
out.lund_coordinates(p4, R)	# F3	primary Lund plane

The same call runs unchanged on CPU and GPU — the backend is chosen from the problem size, not by the caller.

Downstream: the clustering tree as tagger input

Does the tree help a tagger? — setup

The question. Gouskos & Maier [PLuM, 2605.26821] report that explicit Lund-plane splittings help a transformer tagger. We test that on a **full 10-class** training.

Model. Particle Transformer [2202.03772] — 8 encoder + 2 class-attention blocks, 8 heads, embed 128; **2.144 M** parameters.

Data. JetClass, 10 classes, 100 M training jets, **~20.05 M** test jets, 128 particles/jet.

Training. Ranger, lr 10^{-3} , batch 512, 10^6 **iterations** to convergence. All arms read **identical shards** in the same order; features are computed **live** by flashjet inside the loop.

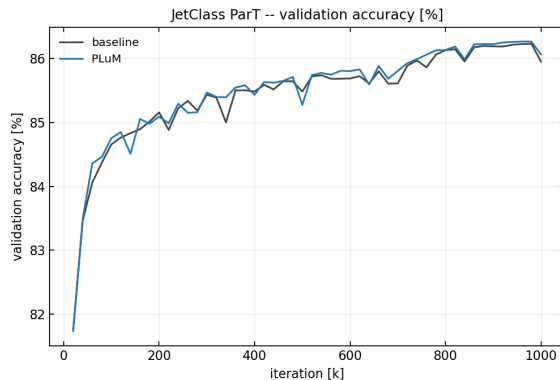
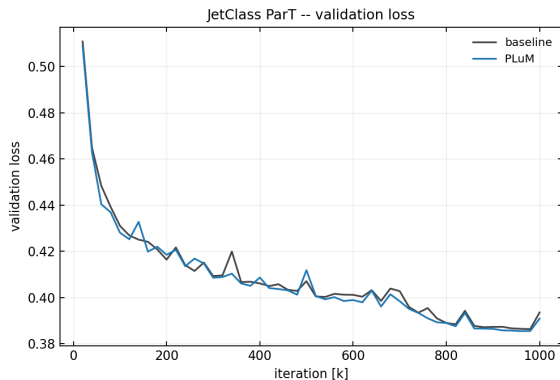
Our baseline reproduces the paper

	ours	published
test accuracy	86.21 %	~86.1 %
parameters	2,193,930	2.19 M

Like-for-like — so a difference between arms is the *feature*, not a broken baseline.

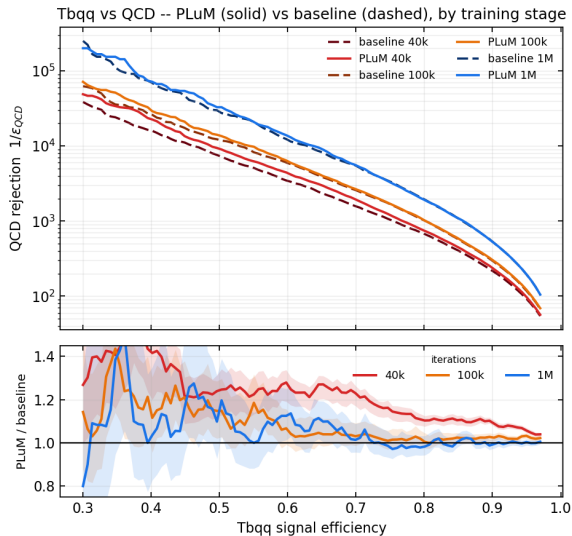
Shown here: PLuM — 48 k_t -ordered Lund splittings $[\ln k_t, \ln 1/\Delta R, \ln z]$, concatenated as extra tokens (176 vs 128). Two other encodings were tried — **subject** (n -body mass, p_T fraction, n_{const}) and **CA5** (per-particle C/A branch points); both null-or-worse (backup). Bootstrap errors, 16–84th pct. **One seed per arm.**

Both arms train identically



Validation loss and accuracy over the full 10^6 iterations. The curves are **on top of each other** — the feature block does not destabilise training, and it does not accelerate it either. Both converge to $\sim 86.2\%$.

Top tagging: the gain closes with training



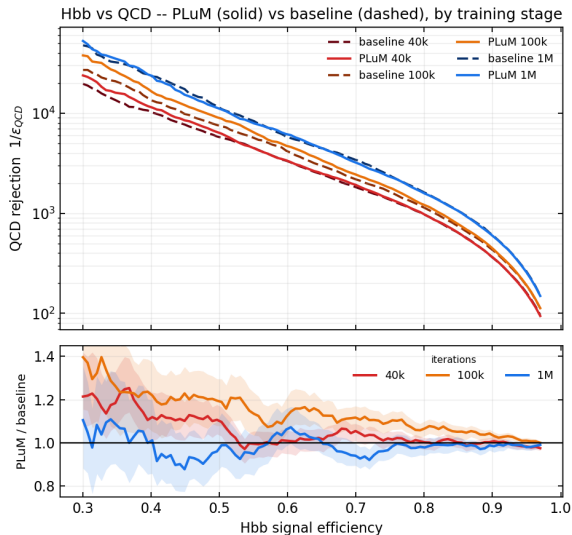
$t \rightarrow bqq$ vs QCD.

Top panel: full ROC at three training stages.
Bottom: **PLuM / baseline** rejection ratio.

At **40k** the ratio sits **10–25 % above 1**. By **1M** it has collapsed onto the line.

Bands: unpaired bootstrap of the test set, 16–84th percentile. The low-efficiency end is noisy because rejection there is $\sim 10^5$.

$H \rightarrow b\bar{b}$: the paper's best channel

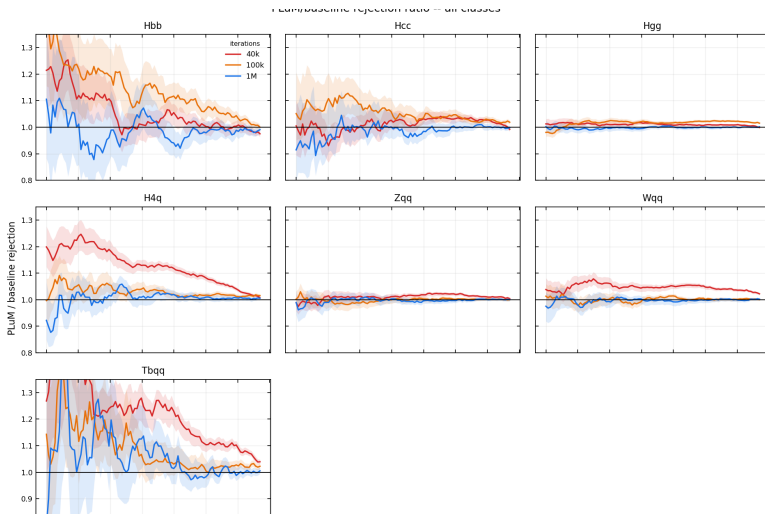


$H \rightarrow b\bar{b}$ vs QCD — where PLuM report their largest gain.

We **reproduce it early**: +20 % at 100k. At **1M** the ratio is **at or below 1**.

Their result is obtained in *binary* mode on a fast simulation **without secondary-vertex inputs**; ours is a 10-class training. The early-training agreement is real — the disagreement is about what survives to convergence.

All seven signal classes — the same story



PLuM/baseline rejection ratio; unpaired-bootstrap bands.

Test accuracy ~ 20.05 M jets

	40k	100k	1M
baseline	83.431	84.632	86.212
PLuM	83.428	84.697	86.206
Δ	-0.003	+0.065	-0.006

What we see

An early gain in every class, gone by 1 M.
Even Wqq — the *fewest* splittings — gains at $\sim 9\sigma$ early: **not confined to complex or *b*-jets**. At convergence: **null-or-worse**.

What we do *not* claim

- **No mechanism** — *b*-specificity excluded.
- **One seed per arm** — a second seed's spread exceeded the effect.
- Their Hbb reproduced at 100k, **not** at convergence.

- 1 **Clusters on the GPU and keeps the full merge history**, so substructure — exclusive subjets, soft drop, Lund coordinates — comes free as array reads. Built for **reclustering inside a training loop**, on tensors that never leave the GPU.
- 2 **Gives the same jets as FastJet**. 150/150 grid points at 100 % n_{jets} agreement, across anti- k_t/k_t /C-A, $R = 0.4\text{--}1.2$ and 24–123 constituents per jet.
- 3 **Reproduces the experiment's own reconstructed jets** exactly, at ~ 600 clusters per event.
- 4 **54–143 $\times$ faster** than vectorised FastJet **in the jet regime** (one large- R jet per timed unit, batched), peaking in the boosted-object range. Cost is independent of R and of algorithm.
- 5 Still a **prototype**, with meaningful headroom left.

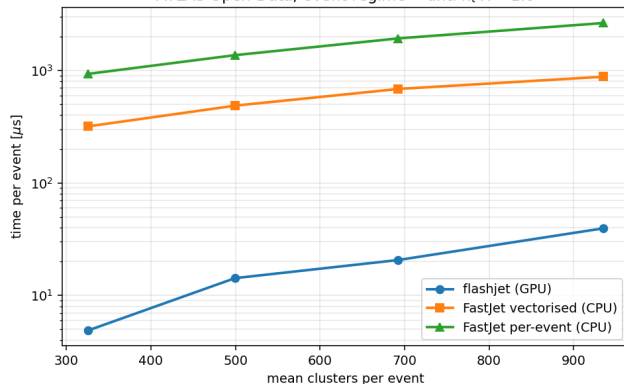
Thank you

Questions?

Benchmarks: ATLAS Open Data (CC0).

Physics closures: analytic predictions on toy showers.

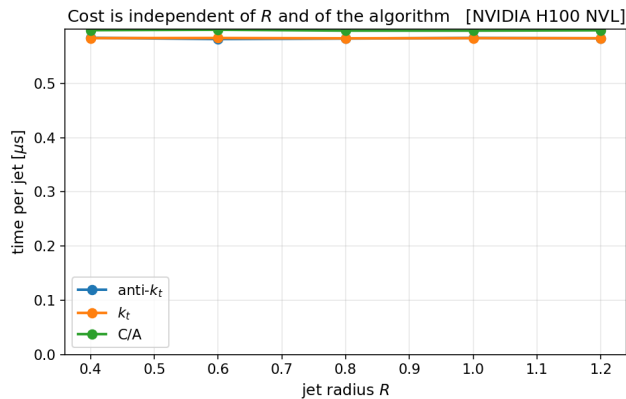
Backup

ATLAS Open Data, event regime -- anti- k_t $R = 1.0$ 

One timed unit = one whole event (~ 590 ATLAS calorimeter clusters), 3 000 events, same method as the jet regime.

$\langle N \rangle$	$\mu\text{s/event}$	speedup
325	4.9	65×
499	14.3	34×
693	20.6	33×
935	39.5	22×

22–65× over vectorised FastJet. Jet-count agreement: **60/75 grid points exactly 100 %**, worst **99.785 %** (busiest bin); leading-jet p_T within 10^{-4} at **all 75**. **H100 NVL** — the same card as the jet-regime plots, so the two are directly comparable.

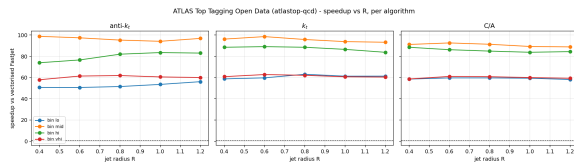
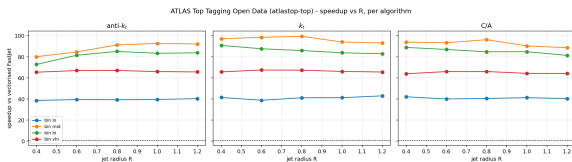


Across $R = 0.4 \rightarrow 1.2$: a **0.3 % spread** over a $3\times$ range in R .

R changes *which* pairs merge, not *how many* distances are computed.

Across algorithms: anti- k_t , k_t and C/A agree to within $\sim 4\%$ at every multiplicity.

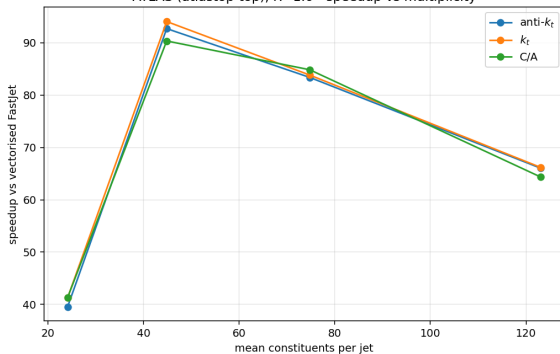
k_t and C/A cost the same as anti- k_t .



boosted top

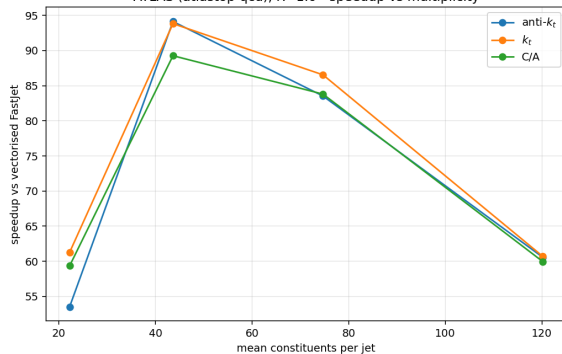
QCD (light q/g)Flat in R for all three algorithms; ordering is by multiplicity bin.

ATLAS (atlastop-top), R=1.0 - speedup vs multiplicity



boosted top

ATLAS (atlastop-qcd), R=1.0 - speedup vs multiplicity



QCD (light q/g)

Backup — the numbers (jet regime)

sample	alg	$\langle n \rangle$	flashjet [$\mu\text{s}/\text{jet}$]	FastJet vec. [$\mu\text{s}/\text{jet}$]	speedup
top	anti- k_t	24.1	0.427	16.5	39×
top	anti- k_t	44.8	0.464	43.0	93×
top	C/A	74.7	1.000	84.8	85×
top	C/A	123.1	2.565	165.1	64×
qcd	anti- k_t	22.2	0.346	17.5	51×
qcd	anti- k_t	43.6	0.463	45.7	99×
qcd	anti- k_t	74.7	1.215	89.8	74×
qcd	anti- k_t	120.2	2.673	154.9	58×

54–143× over vectorised FastJet (median 89×), with 100 % jet-count agreement at all 150 points. **One timed unit = one jet**, batched. Benchmarked on an **NVIDIA H100 NVL**.

$R = 1.0$ rows shown. Against the per-jet FastJet interface the gap is roughly *two orders of magnitude*.

Backup — where the time goes

Nsight Systems — share of GPU time

kernel	% GPU	avg/call
_cluster_large_kernel	97.6	5.47 ms
_decode_kernel	1.5	82 μ s
torch glue	< 1	—

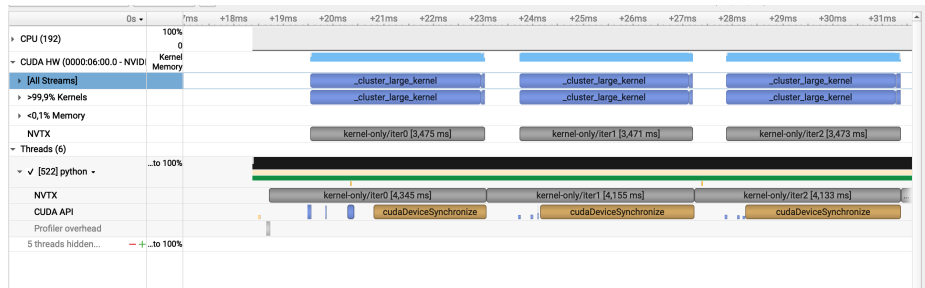
No hidden overhead — essentially all the time is real clustering work, and the memory row of the timeline is empty.

Nsight Compute — hardware counters

metric	value
DRAM throughput	0.02 %
Compute (SM) throughput	15.4 %
Registers / thread	168
Theoretical occupancy	18.75 %
Achieved occupancy	6.25 %
Waves per SM	0.32

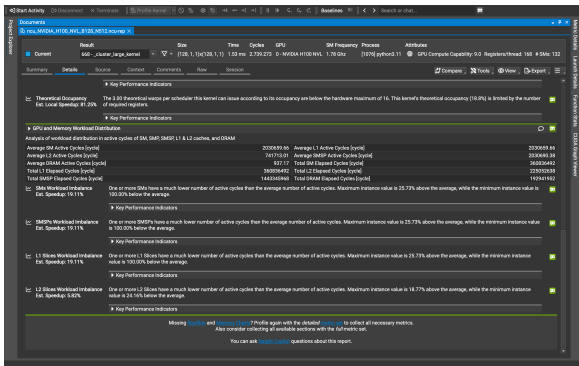
Not memory- or compute-bound — the GPU is simply **not filled**.

Backup — the GPU timeline



Nsight Systems, three consecutive iterations. Each block is one clustering call; the tool labels the rows “>99.9 % Kernels” and “<0.1 % Memory”. The kernels run back to back with nothing in between and the memory row is empty — no host/device shuffling while clustering runs. That is what makes the speedup real rather than an artefact of where the timer was started.

Backup — the profiler names the bottleneck



The tool's own speedup estimates:

finding est. speedup

Theoretical occupancy (registers) 81 %

Achieved occupancy 67 %

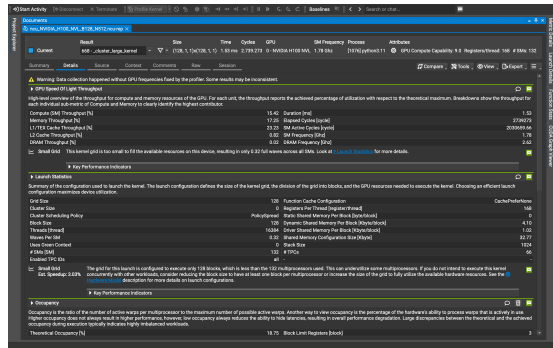
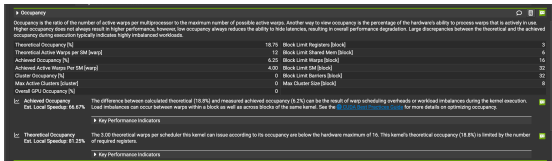
SM workload imbalance 19 %

"This kernel grid is too small to fill the available resources on this device, resulting in only 0.32 full waves across all SMs."

Grid size **128 blocks** on a **132-SM** device — fewer blocks than the GPU has processors.

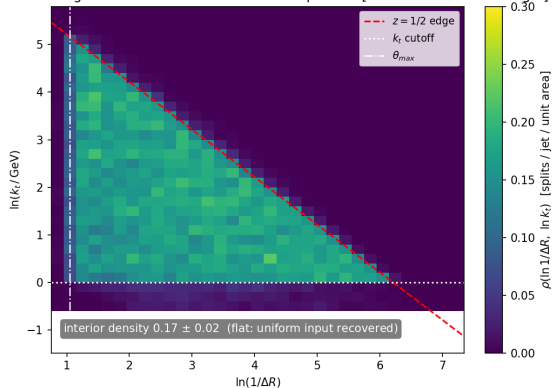
~**1.7–1.8×** more available from launch configuration and register budget — **not** from the algorithm.

Backup — occupancy detail



Register pressure (168/thread) caps theoretical occupancy at 18.75 %; achieved is 6.25 %. With 0.32 waves/SM the kernel does not fill the GPU even once, while DRAM throughput is 0.02 % — this is parallelism starvation, not a bandwidth problem. Both are configuration inherited from smaller GPUs, not properties of the algorithm.

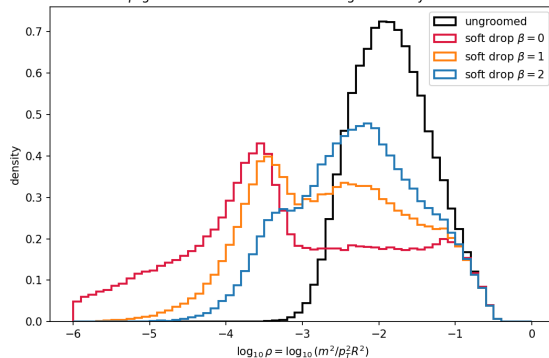
primary Lund plane of the toy shower (C/A, R=0.4):
m-in-triangle emissions come back as a flat plateau [cf. arXiv:1807.04758 Fig 2]



The Lund plane is populated **uniformly** inside the kinematic **triangle** — the leading-log expectation.

Independent NumPy tree-walks pin every decoder, and the d -channel of the Lund output ties *exactly* to the splitting scales computed separately.

Soft-drop β -ordering on REAL QCD jets (164292 jets, $z_{cut} = 0.1$)
smaller β grooms harder — same ordering as the toy-shower closure



The soft-drop β -family orders as predicted: larger β grooms less.

Backup — Input A: the toy fat jets

Purpose. QCD vs W is a *known* truth, so any observable that fails to separate them — or lands off the analytic curve — is a real bug.

Two single-fat-jet samples at $p_T \approx 500$ GeV, $R = 0.8$:

- **QCD-like:** one hard collinear core (92 % of p_T , $\sigma_{y,\phi} = 0.06$) + soft wide-angle radiation (8 %, $\sigma = 0.30$)
- **W-like:** two hard prongs with pair mass $m = \sqrt{z(1-z)} p_T$, $\Delta R = 80.4$ GeV, $z \in [0.30, 0.45]$, random orientation, + 6 % soft contamination

Each “spray” is n massive pions with exponential p_T fractions:

```
frac = rng.exponential(1., n); frac /= frac.sum()
pt    = pt_total * frac
y, phi = rng.normal(y0, sigma, n), rng.normal(phi0, sigma, n)
mt     = np.sqrt(M_PI**2 + pt**2)
```

1500 events per sample. flashjet has **no event generator** — it clusters four-vectors; these toys live in the plot scripts, outside the library.

Backup — Input B: the toy parton shower

A **primary, fixed-coupling, leading-log** shower: emissions sampled **uniformly in the Lund triangle** with density $\bar{\alpha}$ per unit $(\ln 1/\theta, \ln k_t)$ area, off a hard spine.

$$\theta = e^{-u}, \quad k_t = e^v, \quad z = \frac{k_t}{p_{T0} \theta}$$

keep $v \leq \ln(p_{T0}/2) - u$ (i.e. $z \leq \frac{1}{2}$), $k_t > k_t^{\min}$

Why this is the right test

The generator's density is *uniform in the triangle by construction*, so the leading-log predictions — the $1/z$ form of z_g , the β -ordering of the groomed mass — are unambiguous. Landing on them tests the **decoders**, not the shower.

Jet areas use the anti- k_t -paper construction: one hard event plus a dense grid of **infinitely soft ghosts** ($p_T = 10^{-8}$); the ghosts each algorithm sweeps up trace its catchment area.

$\bar{\alpha} = 0.25$, $p_{T0} = 1$ TeV, $R = 0.4$, 20 000 showers; fixed seed, fully re-runnable.

Backup — cost of each feature arm

Wall clock per training step, relative to the baseline:

arm	cost
PLuM	1.59×
subjet	1.42×
CA5	1.22×

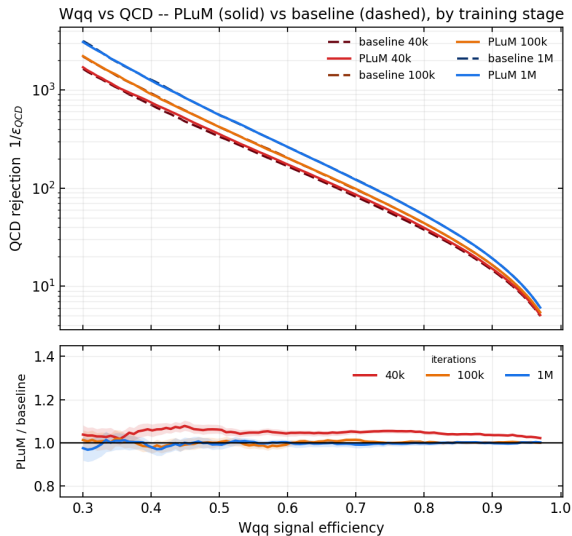
Features computed live by flashjet inside the loop, on the batch already resident on the GPU.

Parameters: 2.144 M baseline $\rightarrow$ 2.194 M with the PLuM feature block (+0.051 M).

Measured before the GPU-side collation change; the clustering itself is a small fraction of a step, so these are upper bounds on the true feature cost.

The cost is the reason a null result still matters: if a feature is free, a null is merely uninteresting — at **1.59×** it is a reason not to use it.

Backup — $W \rightarrow qq$, the fewest-splitting channel



$W \rightarrow qq$ has the **fewest** Lund splittings of any signal class (21.6/jet), yet shows the same early gain.

The effect is **not** confined to complex or b -containing jets — which is what a decay-structure explanation would predict.

Backup — two kinds of validation

First: toys, where truth is known

On generated inputs the *answer* is known analytically — so any observable landing off the predicted curve is a real bug, with nothing to hide behind.

Input A — toy fat jets — QCD-like and W-like fat jets

Input B — toy parton shower — leading-log parton shower

Then: real data, against FastJet

On detector-level jets there is no analytic answer — so the reference is **FastJet on the same events**, and the experiment's own reconstructed jets.

Input C — ATLAS Open Data — large- R jets and full events

The order matters: **toys first** pins the algorithm against a known answer; **real data second** shows it survives detector-level input. A failure at either stage means something different.

Backup — the validation ladder

rung	input	reference	result
unit tests	—	independent NumPy tree-walks	129 pass, incl. GPU paths
anti- k_t jet areas	toy B	the anti- k_t paper's own construction	rigid cones, $\rightarrow \pi R^2$
k_t substructure	toy A	known 2-prong vs QCD truth	clean separation
soft-drop z_g	toy B	leading-log $1/z$ prediction	on the curve
groomed mass vs β	toy B	Soft Drop Figs. 3–4	ordering reproduced
Lund plane	toy A	primary-Lund construction	hard split where predicted
jet-by-jet kinematics	real C	FastJet, same events	150/150 points, 100 % n_{jets}
full-event clustering	real C	the experiment's own jets	100 % match, ~ 600 clusters/ev

Each rung is closed against something independent — an analytic prediction, a separate NumPy implementation, or FastJet itself. No rung is validated against another part of flashjet.

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- F. A. Dreyer, G. P. Salam, G. Soyez, *The Lund Jet Plane*, JHEP **12** (2018) 064, arXiv:1807.04758.
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- A. J. Larkoski, S. Marzani, G. Soyez, J. Thaler, *Soft Drop*, JHEP **05** (2014) 146, arXiv:1402.2657.
- M. Cacciari, G. P. Salam, G. Soyez, *The anti- k_t jet clustering algorithm*, JHEP **04** (2008) 063, arXiv:0802.1189; *FastJet user manual*, Eur. Phys. J. C **72** (2012) 1896, arXiv:1111.6097.
- S. Catani, Yu. L. Dokshitzer, M. H. Seymour, B. R. Webber, Nucl. Phys. B **406** (1993) 187 (k_t); Yu. L. Dokshitzer *et al.*, hep-ph/9707323 and M. Wobisch, T. Wengler, hep-ph/9907280 (Cambridge/Aachen).
- J. de Favereau *et al.*, *DELPHES 3*, JHEP **02** (2014) 057, arXiv:1307.6346.

All benchmark data is **ATLAS Open Data** (CC0), read over the public redirector — no grid certificate required.

dataset	used for
ATLAS Top Tagging Open Data	timing + FastJet agreement
2020 ATLAS Jet Reconstruction	closure vs the experiment's own jets

Constituents are calorimeter clusters, massless. Physics closures use toy showers compared against leading-log analytic predictions, so the prediction being tested is unambiguous.